

Activity Guide — senior / module-09-subtraction-nikhilam

IKS Curriculum

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Activity 1 — Complement Flashcards

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior · **Duration:** 20 min

Purpose

Reinforce the day-by-day concepts of Subtraction — Nikhilam (All from 9) through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

40 flashcards (target – number) pre-made

What students do

In pairs, one student holds up “1000 – 738” (etc.), the other shouts the answer within 3 seconds using the Nikhilam method.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (10 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.

- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Project Brief

Project Brief — Computer Subtraction Explainer

Module: Subtraction — Nikhilam (All from 9) · **Band:** Senior

What you will do

Make a one-page explainer poster: “How does my calculator subtract?” Show two’s complement subtraction in binary using the same principle as Nikhilam.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn’t.
- **Final artefact** (a poster / model / digital deliverable).
- **Written reflection** (500 words (Senior)): what you learned, what surprised you, what you’re still uncertain about.
- **Presentation** (5-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 3 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.